

Joaquín Aldunate

Backend Python & Data Engineer

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Professional Summary

Backend Python and Data Engineer with 5+ years building scalable data pipelines, ETL systems, and multi-country analytics. Specialized in Polars, Spark, Databricks (Asset Bundles, Unity Catalog), and reusable libraries. I leverage generative AI (Claude Code) to explore problems, review architecture, and code, directing my technical capability toward what the client actually needs. Design-thinking approach. Testable code, high-impact delivery. Available remote.

Work Experience

- Sep 2025 – Present **Data Engineer, Numerator (formerly Kantar) – Technology, Remote – Chile**
- Transitioned from contractor to permanent staff (March 2026) after demonstrating sustained technical impact; expanded scope leading multiple parallel projects.
 - Multi-country data workflow design with clients, with multi-agent orchestration (Claude Code) for ETL automation, code review, and portfolio management (Linear/Azure DevOps).
 - Validated that Databricks migrations preserved legacy system behavior, generating synthetic test data from the real schemas.
 - Designed a two-stage automated code review system (generation + independent verification) to reduce manual intervention on repetitive tasks.
- Dec 2024 – Aug 2025 **Data Engineer / Backend Developer, Kantar – Technology, Remote – Chile**
- Development of simulators and multi-country ETL pipelines (5 LATAM countries) with Python, Polars, and PySpark.
 - Databricks architecture: Asset Bundles, Unity Catalog, CI/CD with Azure DevOps.
 - Creation of reusable libraries (storage, merge, simulators) published to Azure Artifacts.
 - Internal dashboard (FastAPI + React) that reduced 200+ man-hours/month in data quality team.
 - Refactoring of notebooks to automated backend with 100+ unit tests (pytest).
- Nov 2023 – Nov 2024 **Tech Lead, eVink – R&D Department, Remote – Chile**
- Led team of 3 developers building backend (Python) and embedded firmware (C++) for wireless network of real-time industrial energy meters.
 - Designed hardware and firmware for miniaturized, scalable, and testable data-loggers.
- Aug 2021 – Oct 2023 **R&D Engineer, Trefimet S.A. – R&D Department, Remote/Hybrid – Chile**
- Led "Smart Furnace Opening" program: firmware for three prototypes + Python backend for real-time remote monitoring and control.
- Mar 2020 – Jul 2021 **R&D Engineer, Komatsu Reman Center Chile, Remote/Hybrid – Chile**
- Developed automated data acquisition and reporting systems in Python for equipment with intermittent connectivity.
 - Designed microservices architecture and machine learning modules for actionable insights extraction.

Education

- 2011 – 2018 **Ph.D. in Engineering Sciences, Electronics and Communications, Pontificia Universidad Católica de Chile, Santiago, Chile**
Thesis: Performance of wireless sensor networks with multi-antenna technology
- 2005 – 2011 **Electrical Engineer (B.Sc. + M.Sc.), Pontificia Universidad Católica de Chile, Santiago, Chile**

Technical Skills

Python	Expert – Polars, Pandas, FastAPI, PySpark, pytest, Typer CLI, packaging (Poetry/Hatch)
Data Engineering	Databricks (Asset Bundles, Unity Catalog), Spark, Delta Lake, advanced SQL, ETL
Frontend	React 18, Vite, TanStack Query, D3.js/Plotly for data visualization
DevOps	Git (worktrees for parallel feature development), Docker, Azure DevOps (CI/CD, Artifacts), Linux
Generative AI	Claude Code, prompt engineering, agentic workflows, LLM-assisted code review, AI-driven CI/CD automation
Soft Skills	Autonomous remote work, technical documentation, process optimization

Languages

- Spanish Native
- English Full professional proficiency (fluent technical reading/writing)